

AP CSP | Lesson 23: Systems & Networks

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5 requires exactly TWO choices. Show paths, calculations and reasons. Treat all networks and timings as the stated classroom models.

1. Which best describes a computing system? [1 mark]

- A. One screen only
- B. Devices and programs working toward a shared purpose
- C. Every file on a disk
- D. Only devices connected to the Internet

Answer: _____

2. What makes S-A-C-T a valid path? [1 mark]

- A. The labels are alphabetical.
- B. It uses the fewest labels possible.
- C. Every consecutive pair has a direct link.
- D. The drawing places S left of T.

Answer: _____

3. In a common name-to-address lookup, what does DNS return? [1 mark]

- A. Address information for the name
- B. The entire requested website
- C. A guaranteed fixed route
- D. The user's password

Answer: _____

4. Which statement about the Web is correct? [1 mark]

- A. It is all of the Internet.
- B. It is the same as Wi-Fi.
- C. It replaces Internet protocols.
- D. It is a service that uses the Internet.

Answer: _____

5. Which TWO statements are correct? [1 mark]

- A. Open protocols support compatible implementations.
- B. A network must contain a website.
- C. Bandwidth is usually measured in bits per second.
- D. More routes automatically mean more bits per second.

Answer: _____

6. Ignore overhead. How long does sending 90 megabits at 30 megabits/s take? [1 mark]

- A. 0.33 s
- B. 3 s
- C. 60 s
- D. 2700 s

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A sensor sends readings to a server, which updates a display. (a) Identify one computing device. (b) State a shared system purpose. (c) Explain what makes their connections a network. (d) Identify input to the server.

8. [4 marks]

Model edges: S-A, S-B, A-C, A-D, B-D, C-T, D-T, C-D, all undirected. (a) Give a valid S-to-T path. (b) Count its hops. (c) Explain why S-B-C-T is invalid. (d) Explain why drawing distance alone does not select a route.

9. [4 marks]

A client looks up a site name and then requests a page. (a) Describe the DNS lookup result. (b) Describe the IP address's role. (c) State what the page response contains. (d) Explain why DNS is not the same as the page server.

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Written responses | Questions 10-12

10. [4 marks]

An 80-megabit payload is sent at an ideal 20 megabits/s. Ignore overhead. (a) Calculate time. (b) Give the bandwidth unit. (c) Would 40 megabits/s halve ideal time? Calculate. (d) Does adding a graph route prove capacity doubled? Explain.

11. [4 marks]

(a) Describe the Internet. (b) Describe the Web. (c) Name the Web protocol taught here. (d) Explain why one unavailable web page does not prove the entire Internet is unavailable.

12. [4 marks]

A new device joins a network. (a) What is a protocol? (b) How can open standards help compatibility? (c) What connection is needed for Internet access? (d) Give one reason common protocols do not guarantee unlimited capacity.
